Which String Vibrations Become Which Generations on the Dwork Quintic

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We determine, for the heterotic string on the one-parameter family of quintic threefolds, which vibration modes become which families of charged particles. The hundred and one families are labelled by the characters of a symmetry group of order one hundred and twenty-five. Every Yukawa coupling that this symmetry allows is a power of the modulus, and every other coupling vanishes. The invariant deformation pairs each family with exactly one partner, so that fifty families are massless at the Fermat point and none anywhere else, and at the Fermat point the masses are given in closed form by the Gamma function. We also prove that the Geometric Selection Conjecture is equivalent to the existence of one viable vacuum fixed by mirror symmetry, and that with the standard embedding no quintic threefold, and no free quotient of one, carries three families.

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I. INTRODUCTION

A closed string moving in ten dimensions has a tower of vibration modes, and the massless ones of the heterotic string on a product of four-dimensional space-time with a compact six-manifold X are the fields of a four-dimensional gauge theory. When X is a Calabi–Yau threefold and the gauge connection is identified with the Levi-Civita connection of its Ricci-flat metric, the gauge group is E_6 and the charged chiral multiplets come in two conjugate representations whose numbers are the two Hodge numbers of X. This was established in 1985 by Candelas, Horowitz, Strominger and Witten [1] and by Witten [2], and the couplings of three charged fields were identified by Strominger [3] with an integral over X that Candelas [4] computed on the quintic threefold. The question in the title is the one that this dictionary leaves open: a vibration mode is a cohomology class on X, a particle generation is an irreducible piece of the four-dimensional spectrum, and one wants to know which class carries which generation, with what mass, and how this changes as X moves in its moduli space.[1]

We answer it completely on one family, the Dwork pencil of quintics

$$X_\psi = \{x_1^5+x_2^5+x_3^5+x_4^5+x_5^5-5\psi x_1x_2x_3x_4x_5=0\}\subset\mathbb{P}^4, \quad (1)$$

smooth for $\psi^5\neq 1$, which is the family on which mirror symmetry was first tested by Candelas, de la Ossa, Green and Parkes [5] and on which the Yukawa couplings were counted by Doran, Greene and Judes [6]. The group $\Gamma\cong(\mathbb{Z}/5)^3$ of diagonal phases preserving the equation acts on every member

[1] The word generation is used here in the sense of the E_6 theory: one copy of the representation **27**, which contains one family of quarks and leptons together with its Higgs and singlet partners. The three observed families are three such copies after the further breaking of E_6; the paper does not touch that breaking, which depends on discrete Wilson lines and on a quotient of X that the quintic does not admit in a form that reduces a hundred and one copies to three, so that the results here concern the full E_6 spectrum of the covering space.

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of the pencil, and the hundred and one classes in $H^{2,1}(X_\psi)$, which are the hundred and one copies of the **27**, are the monomials $x^{\mathbf{b}}$ with exponent vector $\mathbf{b} \in \{0,1,2,3\}^5$ of total degree five. The results are the following.

Theorem 1 (Labels)**.** *The hundred and one classes carry hundred and one distinct characters of* Γ*, each with multiplicity one, so every generation is labelled by its character. Of the hundred and twenty-five characters of* Γ *the twenty-four that do not occur are exactly the classes of the permutations of* $(0,1,2,3,4)$*. The permutation group* S_5 *of the coordinates permutes the labels in five orbits of sizes* 1*,* 20*,* 20*,* 30 *and* 30*, the five types of generation.*

Theorem 2 (Couplings)**.** *Let* $\psi^5 \neq 1$*. The cubic form of the pencil, normalised so that the monomial* $x_1^3 \cdots x_5^3$ *has value one, takes on three generations* $\mathbf{a},\mathbf{b},\mathbf{c}$ *the value* ψ^r *when* $\mathbf{a}+\mathbf{b}+\mathbf{c} \equiv (3-r)\mathbf{1} \pmod 5$ *for some* $r \in \{0,1,2,3\}$*, and zero otherwise. There are* 1431 *unordered triples with a nonzero coupling,* 566*,* 425*,* 280 *and* 160 *of them with* $r = 0,1,2,3$*, and all* 1431 *are nonzero at every smooth member with* $\psi \neq 0$*.*

Theorem 3 (Mass map)**.** *The invariant class* $h = x_1x_2x_3x_4x_5$*, the direction of the pencil itself, pairs each generation* $\mathbf{b}$ *with exactly one generation* $\sigma(\mathbf{b})$*, where* σ *is an involution of the labels with one fixed point. The coupling of* h*,* $\mathbf{b}$ *and* $\sigma(\mathbf{b})$ *equals* 1 *for the fifty-one labels with all entries at most two,* ψ *for the thirty labels of type* $(3,1,1,0,0)$ *and* ψ^2 *for the twenty labels of type* $(3,2,0,0,0)$*. Hence the mass matrix induced by a vacuum value of the* **27** *labelled by* h *has full rank one hundred and one for every* $\psi \neq 0$*,* $\psi^5 \neq 1$*, its determinant is* ψ^{70}*, and at the Fermat point* $\psi = 0$ *its rank is fifty-one with the fifty generations containing an exponent three massless. The spectrum consists of one singlet and fifty doublets, and the* Γ*-invariant metric is diagonal in the labels, so the fifty doublet masses are, in units of the vacuum value,* $|\psi|^{r(\mathbf{b})}\left(g_{\mathbf{b}}\, g_{\sigma(\mathbf{b})}\right)^{-1/2}$ *with* g *the Hodge norms of the classes.*

Theorem 2 is stronger than the count of Doran, Greene and Judes [6], who found by a search that 1431 couplings are allowed by the symmetry; it shows that the symmetry is the only source of zeros, and gives every allowed coupling in closed form. The number 1431 is recovered by hand in Section V A. The theorems are proved in Sections III to VI after the dictionary between vibrations and particles is set out, with complete proofs, in Section II. Section VII evaluates the Hodge norms at the Fermat point in terms of the Gamma function, which turns the mass formula of Theorem 3 into numbers there.

The fourth result concerns the choice of the vacuum. A conjecture of Bhattacharjee, Bhattacharya and Ghosh [24], the Geometric Selection Conjecture, asks for a functional on the heterotic landscape, built from the Weil–Petersson metric, that is proper, has a unique minimum, takes that minimum at a phenomenologically viable vacuum and is invariant under mirror symmetry. The precise statement is recalled in Section VIII.

Theorem 4 (Selection)**.** *The Geometric Selection Conjecture holds if and only if there is a viable vacuum, in the sense of conditions* (C1) *to* (C9) *of [24], that is not moved by the mirror involution. On the Dwork pencil the mass condition* (C8) *holds in the precise form that the mass map of the invariant direction has full rank with entries* 1*,* ψ *and* ψ^2*, at every member with* $\psi \neq 0$*; the three-family condition* (C2) *fails on every smooth quintic and on every quotient of a smooth quintic by a group acting freely.*

The first part says that the functional carries no information beyond the vacuum it selects: once a viable vacuum is given, a functional with all the required properties is written down in Section VIII from the metric alone, and conversely the minimum of any such functional is a viable vacuum. What remains of the conjecture is therefore the existence of one viable vacuum, which is where its content lies. The second part locates the Dwork pencil with respect to that question: its mass spectrum has the full rank and the hierarchy of scales that condition (C8) asks for, and the obstruction is the number of families, a hundred and one on the quintic and $100/|H|$ on a free quotient by a group H, never three.

The constructions behind the four theorems were organised on the Anaxia engine of the BLLM programme [27, 28].[2]

History

The dictionary between the vibrations of the string and the fields in four dimensions on a Calabi–Yau background was written down in [1]; the decomposition of the adjoint of E_8 under $E_6 \times SU(3)$ and the identification of the massless charged fields with the Dolbeault cohomology of the tangent bundle are there, and the identification of the number of generations with half the Euler characteristic. Witten [2] discussed the breaking of E_6 by Wilson lines and the pattern of masses from vacuum values of the **27**, which is the mechanism behind Theorem 3. Strominger [3] identified the cubic couplings of the **27** with the cubic form on $H^1(X, T_X)$ given by the holomorphic three-form, and the normalisation by the Kähler metric of the matter fields was clarified by Dixon, Kaplunovsky and Louis [8] and by

[2] Anaxia runs a failure-driven loop: each strategy for a target is a list of checks, each check is a computation or a proof with a named source, a failed check is recorded together with the change of strategy it forces, and the label of a target (open, candidate, proved) is computed by the engine from the record, never assigned by hand. For this paper the engine ran four targets. For the couplings, the first strategy was the count of couplings allowed by the symmetry, as in [6]; its check that the allowed couplings are nonzero failed, because a symmetry count only bounds the number of nonzero couplings from above, and the change it forced was to evaluate the socle functional of Section IV on every monomial of degree fifteen, which passed. For the mass map the single strategy, the pairing by the invariant class, passed at once. For the Fermat point the engine checked the Gamma-function norms against the constant of Distler and Greene [16]. For the selection conjecture the first strategy was to look for the viable vacuum on the quintic itself; its check of the number of families failed for the quintic and for each of its free quotients, and the change it forced was to separate the functional from the vacuum, which gives Theorem 4; the check of the existence of a viable vacuum then failed on the literature, and that target stays open. The engine proposes no mathematics of its own here: every statement is proved in the text, and the record of the loop is part of the archive of the paper [29].

Candelas and de la Ossa [9]. Candelas [4] computed the couplings on the Fermat quintic through the Jacobian ring, with the Gamma-function normalisation of the Hodge norms at that point; Greene and Plesser [7] introduced the quotient of the Dwork pencil by Γ as the mirror of the quintic, and [5] computed the coupling of h with itself along the pencil, $5/(1-\psi^5)$ in their normalisation, which is the case $\mathbf{a}=\mathbf{b}=\mathbf{c}=h$ of Theorem 2 together with Remark 7. Doran, Greene and Judes [6] counted by computer the couplings not forced to vanish by Γ, found 1431, and observed that the count is close to the total divided by the order of the group. The Jacobian-ring description of $H^{2,1}$ of a hypersurface and of its cubic form is Griffiths' [10], and the perfectness of the residue pairing with the Hessian in the socle is Macaulay's theorem in the form given by Carlson and Griffiths [11]. On the physical side, the holomorphic Yukawa couplings of heterotic line-bundle models were computed by Blesneag, Buchbinder, Candelas and Lukas [14], and the physical, metric-normalised couplings of the quintic at the Fermat point by Butbaia, Mayorga Peña, Tan, Berglund, Hübsch, Jejjala and Mishra [15] with numerical Ricci-flat metrics, whose Gamma-function value for the invariant direction agrees with Section VII. Heterotic standard models with three families on other threefolds were constructed by Braun, He, Ovrut and Pantev [20], Bouchard and Donagi [21] and Anderson, Gray, Lukas and Palti [22, 23]; the question of which classes carry which families is posed there in a form that depends on the choice of bundle, and the present paper answers it in the one case where every piece of the data is explicit. Distler and Greene [16] computed the normalised couplings of the Gepner model at the Fermat point exactly, and Section VII recovers their constant from the Hodge norms. The monodromy and intersection numbers used there go back to Pham [17], Sebastiani and Thom [18] and Sakamoto [19]. Two further aspects of the same compactifications, the suppression of the electroweak scale by singularities of the threefold and fibrations of Calabi–Yau manifolds by circles, are treated in [25, 26].

II. THE DICTIONARY

Throughout, X is a compact Kähler threefold with holonomy exactly $SU(3)$, so that X has a Ricci-flat Kähler metric g by Yau's theorem, a nowhere vanishing holomorphic three-form Ω, and $h^{1,0}=h^{2,0}=0$. The tangent bundle T_X with the Levi-Civita connection is a holomorphic $SU(3)$-bundle with a Hermitian Yang–Mills connection; this is the standard embedding.

A. Spinors and the Dolbeault complex

On a Kähler manifold of complex dimension n with trivial canonical bundle the spinor bundle is $S=\bigoplus_q \Lambda^{0,q}T_X^*$, and the Dirac operator of the Levi-Civita connection is $D=\sqrt{2}(\bar\partial+\bar\partial^*)$, with the chirality grading given by the parity of q (Hitchin [12]). If V is a holomorphic Hermitian bundle with a connection whose curvature is of type $(1,1)$, the Dirac operator twisted by V is $\sqrt{2}(\bar\partial_V+\bar\partial_V^*)$ on $\Lambda^{0,\bullet}\otimes V$, and its kernel is the harmonic representative of $\bigoplus_q H^{0,q}(X,V)$. This is the only input from spin geometry.

B. The charged spectrum

Theorem 5 (Spectrum of the standard embedding)**.** *Let X be as above with the standard embedding. The massless charged chiral multiplets of the $E_8\times E_8$ heterotic string on $\mathbb{R}^{3,1}\times X$ in the observable sector transform under E_6 as*

$$h^{2,1}(X)\ \textit{copies of}\ \mathbf{27}\ \oplus\ h^{1,1}(X)\ \textit{copies of}\ \overline{\mathbf{27}},$$

with the $\mathbf{27}$'s in natural bijection with $H^1(X,T_X)\cong H^{2,1}(X)$ and the $\overline{\mathbf{27}}$'s with $H^1(X,T_X^*)\cong H^{1,1}(X)$. The net number of generations is $|h^{2,1}-h^{1,1}|=\tfrac{1}{2}|\chi(X)|$.

Proof. The ten-dimensional gaugino is a Majorana–Weyl spinor with values in the adjoint of E_8. Under $E_8\supset E_6\times SU(3)$,

$$\mathbf{248}=(\mathbf{78},\mathbf{1})\oplus(\mathbf{1},\mathbf{8})\oplus(\mathbf{27},\mathbf{3})\oplus(\overline{\mathbf{27}},\bar{\mathbf{3}}),$$

and the standard embedding identifies the $SU(3)$ factor with the holonomy group acting on T_X and sends $\mathbf{3}$ to T_X, $\bar{\mathbf{3}}$ to T_X^*. A ten-dimensional Weyl spinor on $\mathbb{R}^{3,1}\times X$ decomposes as a four-dimensional Weyl spinor times a six-dimensional one of correlated chirality, so the four-dimensional massless chiral fermions in the $\mathbf{27}$ are the zero modes of the Dirac operator on X twisted by T_X, counted with their six-dimensional chirality; the $SU(3)$ holonomy makes the twisted operator $\sqrt{2}(\bar\partial+\bar\partial^*)$ on $\Lambda^{0,\bullet}\otimes T_X$, and its chiral zero modes are $H^{0,1}(X,T_X)\oplus H^{0,3}(X,T_X)$ and $H^{0,0}(X,T_X)\oplus H^{0,2}(X,T_X)$ for the two chiralities. Since Ω trivialises $\Lambda^3T_X^*$, contraction with Ω gives $H^{0,q}(X,T_X)\cong H^{2,q}(X)$ and $H^{0,q}(X,T_X^*)=H^{1,q}(X)$. With $h^{2,0}=h^{1,0}=0$ and Serre duality, $H^{0,0}(T_X)=H^{2,0}=0$, $H^{0,3}(T_X)=H^{2,3}=\overline{H^{1,0}}^{\,*}=0$, $H^{0,2}(T_X)=H^{2,2}\cong\overline{H^{1,1}}$, and $H^{0,1}(T_X)=H^{2,1}$. A four-dimensional Weyl fermion in the $\mathbf{27}$ of one chirality and one in the $\overline{\mathbf{27}}$ of the opposite chirality are related by conjugation, so the chiral multiplets of definite four-dimensional chirality are $h^{2,1}$ copies of $\mathbf{27}$ from $H^{0,1}(T_X)$ and $h^{1,1}$ copies of $\overline{\mathbf{27}}$ from $H^{0,2}(T_X)$. The index of the twisted Dirac operator is $h^{2,1}-h^{1,1}$, which equals $-\chi(X)/2$ because $\chi=2(h^{1,1}-h^{2,1})$ on a threefold with $h^{1,0}=h^{2,0}=0$. □

C. The cubic form

Let $V=H^1(X,T_X)$. The composition of cup product and contraction,

$$V\otimes V\otimes V\to H^3(X,\Lambda^3T_X)=H^3(X,K_X^{-1})\cong H^3(X,\mathscr{O}_X)\cong\mathbb{C},$$

where the last two isomorphisms use Ω, is a symmetric trilinear form Y on V, the cubic form of X. In terms of the variation of Hodge structure, with ∇ the Gauss–Manin connection and Ω extended holomorphically over the moduli space,

$$Y(a,b,c)=\int_X \Omega\wedge\nabla_a\nabla_b\nabla_c\Omega, \tag{2}$$

and this is the coupling of three **27**'s [3]. Changing Ω to $f\Omega$ multiplies Y by f^2, so only the cubic form up to a nowhere vanishing holomorphic function is intrinsic, and the physical couplings are Y divided by the square roots of the Hodge norms of the three classes and by $\langle\Omega,\overline{\Omega}\rangle$ [8, 9].

D. Mass terms

A vacuum value v of the scalar in the **27** labelled by a class $h \in V$ in a direction that breaks E_6 gives, through the superpotential $Y(h,\cdot,\cdot)$, a Dirac mass term $vY(h,b,c)$ between the fermions labelled by b and c [2]. The physical mass matrix is

$$M_{bc} = v\,\frac{Y(h,b,c)}{\sqrt{g_b\,g_c}} \tag{3}$$

in an orthogonal basis of V, where $g_b = \langle\omega_b,\overline{\omega_b}\rangle$ is the Hodge norm of the harmonic representative, and the masses are its singular values. We call $Y(h,\cdot,\cdot)$ the mass map of h.

E. Hypersurfaces

For a smooth hypersurface $X = \{F = 0\} \subset \mathbb{P}^4$ of degree five let $J_F = (\partial_1 F,\dots,\partial_5 F)$ be the Jacobian ideal and $R = \mathbb{C}[x_1,\dots,x_5]/J_F$ the Jacobian ring, graded by degree. Griffiths' theorem [10] identifies $H^{3-q,q}_{\mathrm{prim}}(X)$ with $R_{5(q+1)-5}$ through residues: for a homogeneous P of degree $5q$,

$$\omega_P = \mathrm{Res}_X\,\frac{P\,\Omega_0}{F^{q+1}}, \qquad \Omega_0 = \sum_i (-1)^i x_i\,dx_1\wedge\cdots\widehat{dx_i}\cdots\wedge dx_5,$$

so $H^{3,0} = R_0$, $H^{2,1} = R_5$, $H^{1,2} = R_{10}$, $H^{0,3} = R_{15}$, and every class of a quintic threefold is primitive in degree three. The ring R is a graded complete intersection with Hilbert series $(1+t+t^2+t^3)^5$, so $\dim R_5 = 101$, $\dim R_{15} = 1$, and R_{15} is the socle. Macaulay's theorem, as stated in [11], says that the Hessian $\mathrm{Hess}\,F = \det(\partial_i\partial_j F)$ has nonzero image in R_{15} and that the cubic form is multiplication in R followed by the residue functional $\lambda_F : R_{15} \to \mathbb{C}$, $\lambda_F(\mathrm{Hess}\,F) = 1$, up to a constant depending only on the normalisation of Ω:

$$Y(\omega_P,\omega_Q,\omega_S) = c\,\lambda_F(PQS). \tag{4}$$

Figure 1 shows a curve on the Fermat quintic and the Hodge numbers of a quintic threefold: by Theorem 5 the hundred and one classes of type $(2,1)$ are the generations and the single class of type $(1,1)$ is the antigeneration.

III. LABELS

Let $\zeta = e^{2\pi i/5}$ and let $K = \{a \in (\mathbb{Z}/5)^5 : \sum a_i = 0\}$ act on $\mathbb{P}^4$ by $x_i \mapsto \zeta^{a_i}x_i$. The diagonal subgroup D acts trivially on $\mathbb{P}^4$, and $\Gamma = K/D \cong (\mathbb{Z}/5)^3$ preserves $F_\psi = \sum x_i^5 - 5\psi\prod x_i$ for every ψ. The characters of Γ are the characters of K trivial on D, that is the classes

$$\widehat{\Gamma} = \{\mathbf{b} \in (\mathbb{Z}/5)^5 : \textstyle\sum b_i = 0\}/\langle\mathbf{1}\rangle, \qquad |\widehat{\Gamma}| = 125,$$

with $\mathbf{b}$ acting by $a \mapsto \zeta^{a\cdot\mathbf{b}}$. A monomial $x^{\mathbf{b}}$ transforms under $a \in K$ by $\zeta^{a\cdot\mathbf{b}}$, so it lies in the isotypic component of the class of $\mathbf{b}$.

Let B be the set of $\mathbf{b} \in \{0,1,2,3\}^5$ with $|\mathbf{b}| = \sum b_i = 5$. The monomials $x^{\mathbf{b}}$, $\mathbf{b} \in B$, are a basis of R_5 for $\psi^5 \neq 1$, because the Jacobian ideal of F_ψ is generated by $x_i^4 - \psi\prod_{j\neq i}x_j$, and modulo these every monomial reduces to one with all exponents at most three; the reduced monomials of degree five are the 101 elements of B, which matches $\dim R_5$. The count is the number of compositions of five into five parts, $\binom{9}{4} = 126$, less the twenty with a part equal to four and the five with a part equal to five.

Proof of Theorem 1. Two elements $\mathbf{b},\mathbf{b}' \in B$ define the same character when $\mathbf{b}' \equiv \mathbf{b} + k\mathbf{1} \pmod 5$ for some k. Since the entries of $\mathbf{b}'$ lie in $\{0,\dots,4\}$, $\mathbf{b}'$ is the vector of residues of $\mathbf{b}+k\mathbf{1}$ in that range: each b'_i is b_i+k or b_i+k-5, and $\sum b'_i = 5$. For $k=1$ the residues are $b_i+1 \le 4$ unless $b_i = 4$, impossible, so all residues are b_i+1, with sum $10 \neq 5$. For $k=2$ the residues are b_i+2 if $b_i \le 2$ and b_i-3 if $b_i = 3$; the sum is $15-5m$ where m is the number of threes, so $m=2$, but two threes already sum to six. For $k=3$ the residues are b_i+3 if $b_i \le 1$ and b_i-2 if $b_i \ge 2$, sum $20-5m'$ with m' the number of entries at least two, so $m'=3$, three entries at least two summing to at most five, impossible. For $k=4$ the residues are b_i+4 if $b_i = 0$ and b_i-1 otherwise, sum $25-5m''$ with m'' the number of nonzero entries, so $m''=4$: four nonzero entries summing to five means $(2,1,1,1,0)$ up to order, and then $\mathbf{b}' = (1,0,0,0,4)$ has a four. So only $k=0$ works and the 101 characters are distinct; being characters of an abelian group acting on R_5, they are then the full decomposition, each isotypic component being one-dimensional.

The group $\widehat{\Gamma}$ has 125 elements and B realises 101 of them, so exactly 24 are missed. A permutation of $(0,1,2,3,4)$ has sum ten, so its class lies in $\widehat{\Gamma}$, and adding $\mathbf{1}$ to it gives another permutation of $(0,1,2,3,4)$ modulo five; the hundred and twenty permutations therefore fall into 24 classes of five. None of these classes meets B, because every representative with entries in $\{0,\dots,4\}$ is a permutation of $(0,1,2,3,4)$ and contains a four. Hence the 24 missed classes are exactly the classes of the permutations.

Finally S_5 permutes the exponent vectors, and the orbits in B are the multisets of entries: $\{1,1,1,1,1\}$ (one element), $\{2,1,1,1,0\}$ (twenty), $\{3,2,0,0,0\}$ (twenty), $\{2,2,1,0,0\}$ (thirty) and $\{3,1,1,0,0\}$ (thirty). □

Figure 2 shows the hundred and one labels.

IV. THE SOCLE FUNCTIONAL

Fix ψ with $\psi^5 \neq 1$, write $R = R_{F_\psi}$ and $m_0 = x_1^3x_2^3x_3^3x_4^3x_5^3$, the only monomial of degree fifteen with all exponents at most three. Define the sector of a monomial $x^{\mathbf{e}}$ of degree fifteen as follows: $\mathbf{e}$ is in sector $r \in \{0,1,2,3,4\}$ if $e_i \equiv 3-r \pmod 5$ for every i; otherwise $\mathbf{e}$ is in no sector. Sector 4 is empty, because its exponents are at least four each.

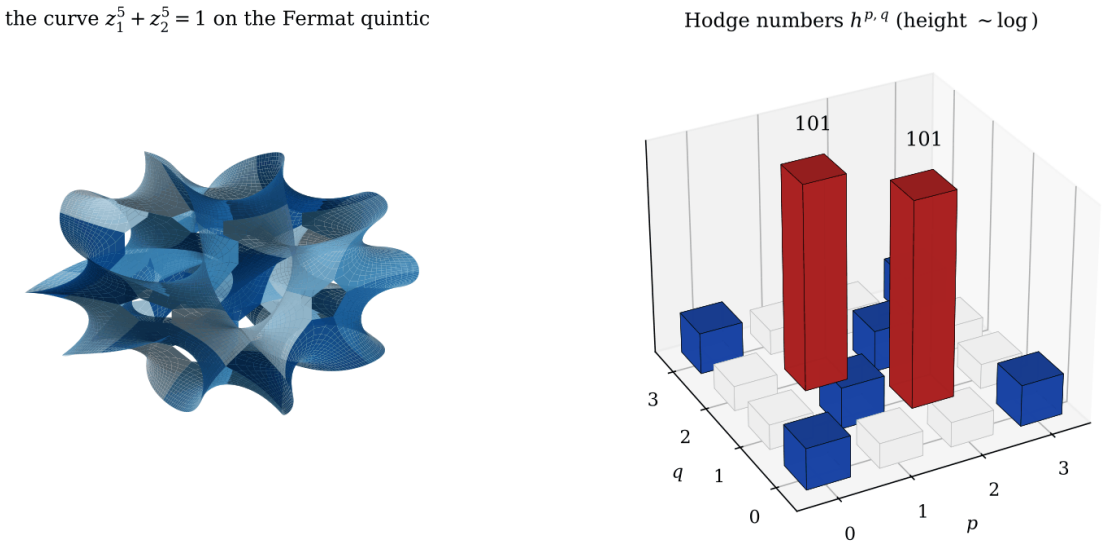


FIG. 1. Left: the intersection of the Fermat quintic with the plane $x_3 = x_4 = 0$, which in an affine chart is the curve $z_1^5 + z_2^5 = 1$ in $\mathbb{C}^2$, drawn with the parametrisation of Hanson [13] and projected to three dimensions along $(\mathrm{Re}\, z_1, \mathrm{Re}\, z_2, \cos\alpha\, \mathrm{Im}\, z_1 + \sin\alpha\, \mathrm{Im}\, z_2)$. It is assembled from twenty-five patches, one for each pair of fifth roots of unity; Γ preserves the curve and acts on it through $\mu_5 \times \mu_5$, permuting the patches. Right: the Hodge numbers $h^{p,q}$ of a smooth quintic threefold, with height proportional to $1 + \log_2 h^{p,q}$; the six blue bars have height one, the two red bars are $h^{2,1} = h^{1,2} = 101$, and the grey cells are zero.

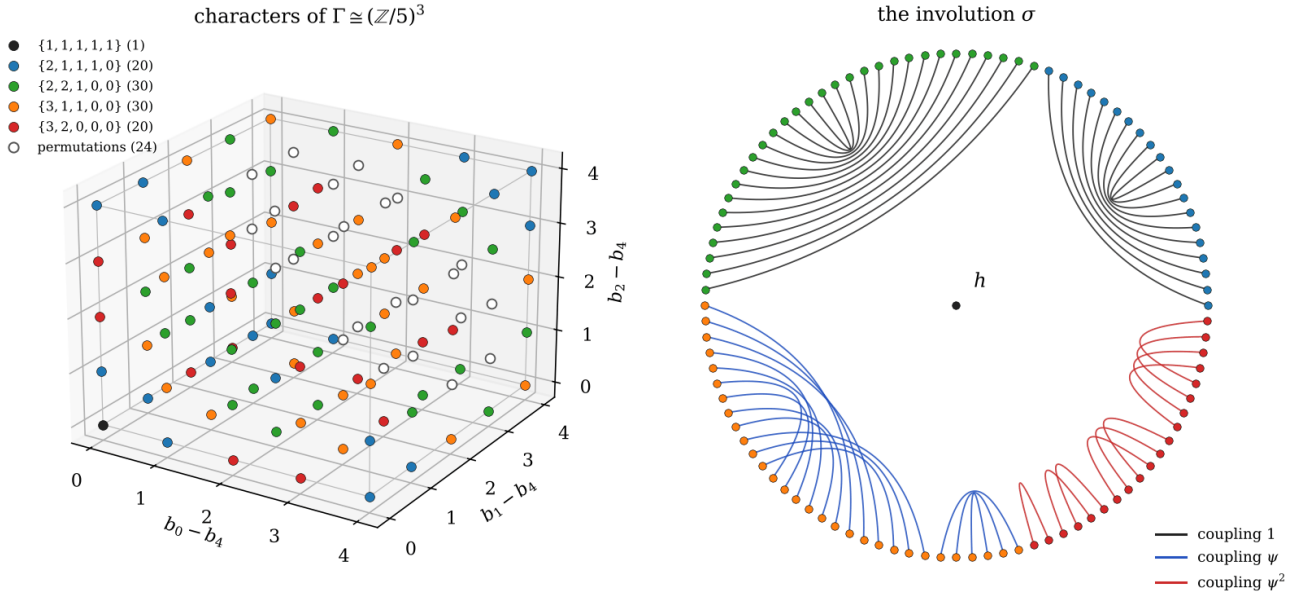


FIG. 2. The hundred and one generations of the Dwork quintic as points of the character group of Γ. Left: $\widehat{\Gamma} \cong (\mathbb{Z}/5)^3$ drawn as a cube of 125 points, the class of $\mathbf{b}$ placed at $(b_0 - b_4, b_1 - b_4, b_2 - b_4)$ mod 5; the 101 labels are filled and coloured by type (the five S_5-orbits), and the 24 classes of permutations of $(0,1,2,3,4)$ are hollow. Right: the involution σ of Theorem 3, drawn as chords between paired labels, with the chord colour giving the power of ψ of the pairing: black for 1, blue for ψ, red for ψ^2; the fixed label is the invariant class h at the centre.

Theorem 6 (Socle functional)**.** *For $\psi^5 \neq 1$, $m_0 \neq 0$ in R_{15}, and the linear functional λ on R_{15} with $\lambda(m_0) = 1$ satisfies, for every monomial $x^{\mathbf{e}}$ of degree fifteen,*

$$\lambda(x^{\mathbf{e}}) = \begin{cases} \psi^r & \text{if } \mathbf{e} \text{ is in sector } r, \\ 0 & \text{otherwise.} \end{cases}$$

Moreover $\lambda(\mathrm{Hess}\, F_\psi) = 3\,200\,000\,(1 - \psi^5)$.

Proof. The relations $x_i^4 \equiv \psi \prod_{j \neq i} x_j$ in R give, for every exponent vector $\mathbf{e}$ with $e_i \geq 4$,

$$x^{\mathbf{e}} \equiv \psi\, x^{\mathbf{e}'}, \qquad \mathbf{e}' = \mathbf{e} + \mathbf{1} - 5\,\mathbf{e}_i, \tag{5}$$

where $\mathbf{e}_i$ is the i-th unit vector. If $\mathbf{e}$ is in sector r then $\mathbf{e}'$ is in

sector $r-1$: its i-th entry is $e_i-4\equiv 3-(r-1)$ and the other entries are $e_j+1\equiv 3-(r-1)$. The sectors are listed by hand. In sector 0 the entries are congruent to 3 and sum to fifteen, so they are all equal to three and $\mathbf{e}=(3,3,3,3,3)$. In sector 1 the entries are congruent to two, so each is 2, 7 or 12; a twelve would leave three for four entries of at least two, so the entries are 2 or 7, and a total of fifteen forces exactly one seven: $\mathbf{e}$ is a permutation of $(7,2,2,2,2)$, and (5) applied to the seven gives ψm_0. In sector 2 the entries are 1, 6 or 11, and the total forces two sixes and three ones (an eleven would leave four for the other four entries, which are congruent to one and so are all one, a total of fifteen; but then $\mathbf{e}=(11,1,1,1,1)$, and (5) applied to the eleven gives $(7,2,2,2,2)$, in sector 1 as well). In sector 3 the entries are multiples of five, so $\mathbf{e}$ is a permutation of $(15,0,0,0,0)$, $(10,5,0,0,0)$ or $(5,5,5,0,0)$, and one reduction at an entry of at least five lands in sector 2. By induction on r, every monomial in sector r satisfies $x^{\mathbf{e}}\equiv\psi^r m_0$ in R.[3]

If $\mathbf{e}$ is in no sector, the monomial $x^{\mathbf{e}}$ has a nontrivial character of Γ: its character is the class of $\mathbf{e}$ modulo $\mathbf{1}$, which is trivial exactly when $\mathbf{e}\equiv c\mathbf{1}$ for some c, that is, when $\mathbf{e}$ is in a sector. The ideal J_{F_ψ} is Γ-stable, so R is graded by the characters of Γ and $x^{\mathbf{e}}$ lies in a nontrivial isotypic component of R_{15}. Since R_{15} is one-dimensional it carries a single character, and m_0 carries the trivial one; so if $m_0\neq 0$, every nontrivial component vanishes and $x^{\mathbf{e}}\equiv 0$.

It remains to show that $m_0\neq 0$. The Hessian $\operatorname{Hess}F_\psi$ has degree fifteen and is Γ-invariant, so its monomials in no sector vanish in R and the others are multiples of m_0: $\operatorname{Hess}F_\psi\equiv H(\psi)m_0$, with $H(\psi)$ the sum over the sector monomials of the Hessian of their coefficients times ψ^r. By Macaulay's theorem $\operatorname{Hess}F_\psi\neq 0$ in R_{15}, so $m_0\neq 0$ and $H(\psi)\neq 0$. The Hessian matrix has diagonal entries $20x_i^3$ and off-diagonal entries $-5\psi\prod_{k\neq i,j}x_k$, each of degree three; an off-diagonal entry omits exactly the two coordinates of its row and column. The determinant is the sum over permutations π of the sign of π times $\prod_i(\text{entry }(i,\pi(i)))$, and the monomial of π depends only on its cycle type: a fixed point a contributes x_a^3 from the diagonal and x_a from every off-diagonal factor, and a moved coordinate m contributes x_m from every off-diagonal factor whose row and column both differ from m. Table I lists the seven cycle types; the derivation of each row is in Appendix A. Summing the last column,

$$\begin{aligned}H(\psi)&=3\,200\,000-\big(2\,000\,000+1\,000\,000+375\,000\\&\qquad+75\,000-187\,500-62\,500\big)\psi^5\\&=3\,200\,000\,(1-\psi^5).\end{aligned}$$

□

Remark 7 (Normalisation). With the residue normalisation $\lambda_F(\operatorname{Hess}F)=1$ of (4), the cubic form of the pencil is $\lambda_{F_\psi}=\lambda/(3\,200\,000(1-\psi^5))$. For $\mathbf{a}=\mathbf{b}=\mathbf{c}=h$ the monomial is $x^{3\mathbf{1}}=m_0$, so $Y(h,h,h)$ is a constant times $1/(1-\psi^5)$, which is the coupling $5/(1-\psi^5)$ of [5] in the gauge in which Ω is the residue of Ω_0/F_ψ; the factor five is the constant c of (4) in that gauge. All ratios of couplings at fixed ψ are independent of these constants, and Theorem 2 is stated for λ.

[3] The reduction (5) uses only the relation $\partial_i F_\psi/5=x_i^4-\psi\prod_{j\neq i}x_j$. It lowers the i-th exponent by four and raises each of the other four by one, so it preserves the degree and moves the class of $\mathbf{e}$ modulo $\mathbf{1}$ by $-5\mathbf{e}_i+\mathbf{1}\equiv\mathbf{1}$; this is why the sectors, which are the classes $(3-r)\mathbf{1}$, are permuted cyclically and why each reduction costs exactly one factor of ψ.

TABLE I. The Hessian of F_ψ in the socle, by cycle type of the permutation. Each row gives the number of permutations, their sign, the monomial they contribute, its sector r, and the total contribution to $H(\psi)$ after the reduction $x^{\mathbf{e}}\equiv\psi^r m_0$.

type	count	sign	monomial	r	contribution
1^5	1	+	$(3,3,3,3,3)$	0	$3\,200\,000$
$2\,1^3$	10	−	$(5,5,5,0,0)$	3	$-2\,000\,000\,\psi^5$
$3\,1^2$	20	+	$(6,6,1,1,1)$	2	$-1\,000\,000\,\psi^5$
$4\,1$	30	−	$(7,2,2,2,2)$	1	$-375\,000\,\psi^5$
5	24	+	$(3,3,3,3,3)$	0	$-75\,000\,\psi^5$
$2^2\,1$	15	+	$(7,2,2,2,2)$	1	$187\,500\,\psi^5$
$3\,2$	20	−	$(3,3,3,3,3)$	0	$62\,500\,\psi^5$

V. COUPLINGS

Proof of Theorem 2. By (4) and Theorem 6, $Y(x^{\mathbf{a}},x^{\mathbf{b}},x^{\mathbf{c}})=\lambda(x^{\mathbf{a}+\mathbf{b}+\mathbf{c}})$ up to the normalisation of Remark 7, and $\mathbf{a}+\mathbf{b}+\mathbf{c}$ has degree fifteen; the value is ψ^r if $\mathbf{a}+\mathbf{b}+\mathbf{c}\equiv(3-r)\mathbf{1}$ and zero otherwise. For $\psi\neq 0$ every sector value is nonzero. The count of the unordered triples is in Section V A. □

A. The count

A triple $\{\mathbf{a},\mathbf{b},\mathbf{c}\}$ of elements of B, with repetitions allowed, has a nonzero coupling exactly when $\mathbf{a}+\mathbf{b}+\mathbf{c}\equiv c\mathbf{1}\pmod 5$ for some c, and then $c\equiv 3-r$ with r the sector. We count ordered triples first. The congruence says that the character of $\mathbf{c}$ is the inverse of the character of $\mathbf{a}+\mathbf{b}$ in $\widehat{\Gamma}$. Every character occurs in B at most once (Theorem 1), so for each ordered pair $(\mathbf{a},\mathbf{b})$ there is at most one $\mathbf{c}$, and there is one exactly when the class of $-(\mathbf{a}+\mathbf{b})$ is one of the hundred and one realised classes, that is, when it is not the class of a permutation of $(0,1,2,3,4)$. The negative of a permutation of $(0,1,2,3,4)$ is, modulo five, a permutation of $(0,1,2,3,4)$ again, so this fails exactly when $\mathbf{a}+\mathbf{b}$ has five distinct residues modulo five.

Excluded pairs. The entries of $\mathbf{a}+\mathbf{b}$ lie in $\{0,\dots,6\}$ and sum to ten. Five distinct residues are the residues $0,1,2,3,4$, whose sum is already ten, so no entry can exceed four and $\mathbf{a}+\mathbf{b}$ is a permutation of $(0,1,2,3,4)$. Fix the permutation, so that the coordinate k of $\mathbf{a}+\mathbf{b}$ equals k for $k=0,\dots,4$ after renaming. Write p_k for the part given to $\mathbf{a}$ in coordinate k; then $0\le p_k\le\min(k,3)$ and $k-p_k\le 3$, so $p_0=0$, $p_1\in\{0,1\}$, $p_2\in\{0,1,2\}$, $p_3\in\{0,1,2,3\}$ and $p_4\in\{1,2,3\}$, and $|\mathbf{a}|=5$ means $p_1+p_2+p_3+p_4=5$. The number of solutions is the coefficient of t^5 in

$$(1+t)(1+t+t^2)(1+t+t^2+t^3)(t+t^2+t^3),$$

which is 16.[4] With 120 permutations there are 1920 excluded ordered pairs, so the number of ordered triples with a nonzero coupling is $101^2 - 1920 = 8281$.

Repetitions. An ordered triple $(\mathbf{a},\mathbf{a},\mathbf{a})$ has a nonzero coupling when $3\mathbf{a} \equiv c\mathbf{1}$, that is, when all entries of $\mathbf{a}$ are congruent modulo five; entries in $\{0,1,2,3\}$ summing to five then force $\mathbf{a} = \mathbf{1}$. For a triple $(\mathbf{a},\mathbf{a},\mathbf{c})$ the partner $\mathbf{c}$ exists unless $2\mathbf{a}$ has five distinct residues; the residues of $2a_i$ for $a_i = 0,1,2,3$ are $0,2,4,1$, and five distinct residues would need five distinct entries of $\mathbf{a}$ in a set of four values, which is impossible. So every $\mathbf{a}$ has a partner $\mathbf{c}$, and $\mathbf{c} = \mathbf{a}$ only for $\mathbf{a} = \mathbf{1}$. There are therefore 100 unordered triples of the form $\{\mathbf{a},\mathbf{a},\mathbf{c}\}$ with $\mathbf{c} \neq \mathbf{a}$, giving 300 ordered triples, one triple $\{\mathbf{1},\mathbf{1},\mathbf{1}\}$, and $8281 - 300 - 1 = 7980$ ordered triples with three distinct entries. The number of unordered triples is

$$\frac{7980}{6} + 100 + 1 = 1330 + 101 = 1431.$$

Sectors. In sector r the sum $\mathbf{e} = \mathbf{a}+\mathbf{b}+\mathbf{c}$ is one of the vectors listed in the proof of Theorem 6 with entries at most nine: $(3,3,3,3,3)$ for $r = 0$, the five permutations of $(7,2,2,2,2)$ for $r = 1$, the ten permutations of $(6,6,1,1,1)$ for $r = 2$ and the ten permutations of $(5,5,5,0,0)$ for $r = 3$. The number $O(\mathbf{e})$ of ordered triples with sum $\mathbf{e}$ is the coefficient of x^5y^5 in $\prod_{i=1}^5 P_{e_i}(x,y)$, where $P_e(x,y) = \sum x^a y^b$ over the pairs $0 \le a,b \le 3$ with $0 \le e-a-b \le 3$. These coefficients are

$$O(3,3,3,3,3) = 3391, \quad O(7,2,2,2,2) = 480,$$
$$O(6,6,1,1,1) = 162, \quad O(5,5,5,0,0) = 87,$$

so the ordered counts by sector are 3391, 2400, 1620 and 870, with total 8281. The hundred unordered triples $\{\mathbf{a},\mathbf{a},\mathbf{c}\}$ with $\mathbf{c} \neq \mathbf{a}$ split as 0, 50, 20 and 30 by sector (the sector of $2\mathbf{a}+\mathbf{c}$), and $\{\mathbf{1},\mathbf{1},\mathbf{1}\}$ lies in sector 0, so the unordered counts are

$$N_r = \frac{O_r - 3D_r - T_r}{6} + D_r + T_r = 566,\ 425,\ 280,\ 160,$$

with O_r, D_r, T_r the ordered, repeated and constant counts of sector r. The coefficient extractions are written out in Appendix B.

VI. THE MASS MAP

Proof of Theorem 3. For $\mathbf{b} \in B$ the coupling $\lambda(x^{\mathbf{b}+\mathbf{c}+\mathbf{1}})$ is nonzero exactly when $\mathbf{b}+\mathbf{c}+\mathbf{1} \equiv (3-r)\mathbf{1}$, that is $\mathbf{c} \equiv (2-r)\mathbf{1} - \mathbf{b} \pmod 5$, with $\mathbf{c} \in B$. For each r the residues of $(2-r)\mathbf{1}-\mathbf{b}$ are determined, and $\mathbf{c}$ exists in B exactly when those residues, taken in $\{0,1,2,3\}$ (a residue 4 excludes), sum to five. Go through the types. If all $b_i \le 2$ then $r = 0$ gives $\mathbf{c} = 2\mathbf{1} - \mathbf{b} \in \{0,1,2\}^5$ of sum five: this is the partner. For $r = 1$ the residues of $\mathbf{1}-\mathbf{b}$ are $1,0,4$ for $b_i = 0,1,2$, so a two in $\mathbf{b}$ excludes, and $\mathbf{b} \in \{0,1\}^5$ of sum five means $\mathbf{b} = \mathbf{1}$, whose $r = 1$ residues are all zero, sum zero: excluded. For $r = 2$ the residues of $-\mathbf{b}$ are $0,4,3$: a one excludes, and $\mathbf{b} \in \{0,2\}^5$ has even sum: excluded. For $r = 3$ the residues of $-\mathbf{1}-\mathbf{b}$ are $4,3,2$: a zero excludes, so $\mathbf{b} \in \{1,2\}^5$ of sum five means $\mathbf{b} = \mathbf{1}$, residues all 3, sum fifteen: excluded. So a label with entries at most two has exactly one partner, $2\mathbf{1}-\mathbf{b}$, with coupling 1; this covers the fifty-one labels of the types $\{1,1,1,1,1\}$, $\{2,1,1,1,0\}$ and $\{2,2,1,0,0\}$, and $\sigma(\mathbf{b}) = 2\mathbf{1}-\mathbf{b}$ is of the same type, with the single fixed point $\mathbf{b} = \mathbf{1}$.

If $\mathbf{b}$ has type $\{3,1,1,0,0\}$: $r = 0$ gives residue 4 at the three, excluded; $r = 1$ gives residues $3,0,0,1,1$, sum five, so $\mathbf{c}$ is the label with 3 where $\mathbf{b}$ has 3, zeros where $\mathbf{b}$ has ones and ones where $\mathbf{b}$ has zeros, of the same type, with coupling ψ; $r = 2$ gives residues $2,4,\ldots$, excluded by the ones; $r = 3$ gives residues $1,3,3,4,4$, excluded. If $\mathbf{b}$ has type $\{3,2,0,0,0\}$: $r = 0$ excluded by the three; $r = 1$ gives residues $3,4,\ldots$, excluded by the two; $r = 2$ gives residues $2,3,0,0,0$, sum five, so $\mathbf{c}$ swaps the three and the two of $\mathbf{b}$, of the same type, coupling ψ^2; $r = 3$ gives residues $1,2,4,4,4$, excluded. This proves that every label has exactly one partner, that σ is an involution preserving type with the one fixed point $\mathbf{1}$, and the three values of the coupling. The matrix $M = (\lambda(x^{\mathbf{b}+\mathbf{c}+\mathbf{1}}))_{\mathbf{b},\mathbf{c}}$ is therefore a monomial matrix with entries 1 (fifty-one times), ψ (thirty times) and ψ^2 (twenty times), so its determinant is $\psi^{30+40} = \psi^{70}$, the sign of σ, a product of fifty transpositions, being $+1$, and its rank is 101 for $\psi \neq 0$. At $\psi = 0$ the rows of the fifty labels containing a three vanish and the remaining fifty-one rows are a permutation matrix on the fifty-one labels with entries at most two: rank fifty-one.

The Γ-invariant Hermitian metric on $H^{2,1}$, the Hodge metric, pairs the isotypic components of distinct characters to zero because a unitary representation of a finite abelian group is an orthogonal sum of its isotypic components; by Theorem 1 the components are the lines spanned by the $x^{\mathbf{b}}$, so the metric is diagonal with entries $g_{\mathbf{b}}$. The mass matrix (3) is then v times a monomial matrix with entries $\psi^{r(\mathbf{b})}(g_{\mathbf{b}}g_{\sigma\mathbf{b}})^{-1/2}$ at $(\mathbf{b},\sigma\mathbf{b})$. The singular values of a symmetric monomial matrix with an involution are the absolute values of its entries, each two-cycle giving a doubly degenerate pair and the fixed point a singlet; this is the stated spectrum. □

Remark 8 (Degenerations). The determinant $\pm\psi^{70}$ shows that the mass map of the invariant direction degenerates on the pencil only at the Fermat point, where the group of automorphisms of X_ψ jumps from $\Gamma \rtimes S_5$ to the full group of order $5^4 \cdot 120$ of the Fermat quintic. The fifty massless generations at $\psi = 0$ are the labels with an exponent three. The larger group acts on a monomial through its full exponent vector, not only through its class modulo $\mathbf{1}$, so at $\psi = 0$ a coupling of h, $\mathbf{b}$ and $\mathbf{c}$ survives only if $\mathbf{1}+\mathbf{b}+\mathbf{c} = (3,3,3,3,3)$; for a label with an exponent three the required partner $2\mathbf{1}-\mathbf{b}$ has a negative entry. At the conifold points $\psi^5 = 1$ the matrix M has nonzero determinant, but the normalisation of Remark 7 has a pole: the physical couplings diverge as $1/(1-\psi^5)$ in the gauge of Ω chosen there, and the Hodge norm of h diverges logarithmically, as in [5].

[4] Expand $(1+t)(1+t+t^2) = 1+2t+2t^2+t^3$ and $(1+t+t^2+t^3)(t+t^2+t^3) = t+2t^2+3t^3+3t^4+2t^5+t^6$. The coefficient of t^5 in the product is $1\cdot2+2\cdot3+2\cdot3+1\cdot2 = 16$, pairing the terms of degree $0,1,2,3$ of the first factor with those of degree $5,4,3,2$ of the second.

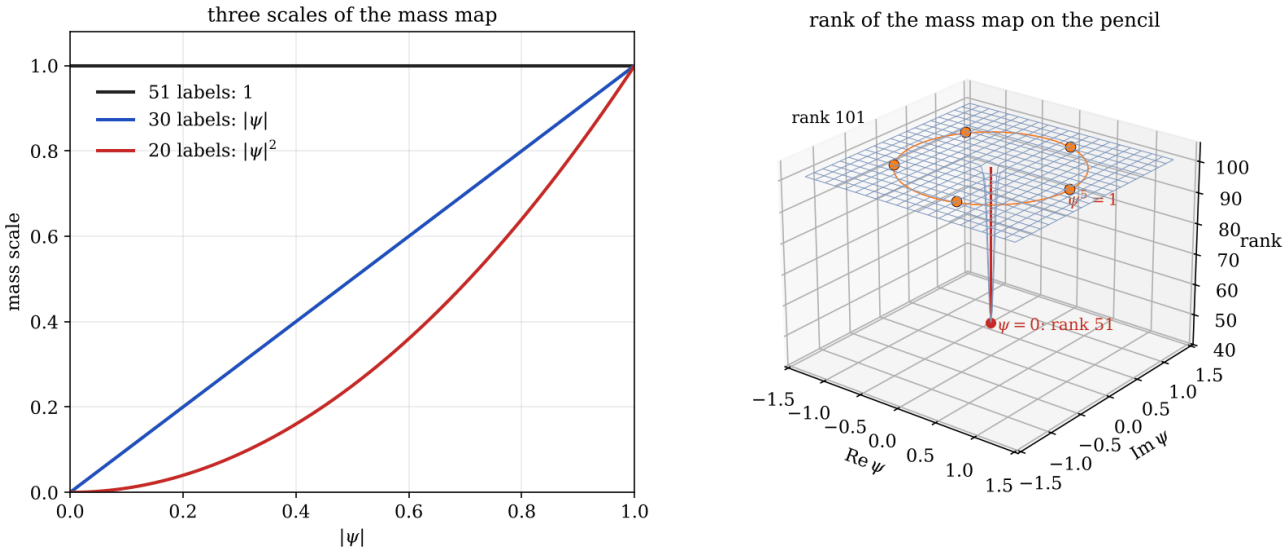


FIG. 3. The mass spectrum of Theorem 3 along the pencil. Left: the three mass scales 1, $|\psi|$ and $|\psi|^2$ against $|\psi|$ for $0 \le |\psi| < 1$, with the multiplicities of the labels (fifty-one, thirty, twenty) that carry them; the singlet h sits in the first group. Right: the rank of the mass map over the ψ-plane, fifty-one at the Fermat point and a hundred and one elsewhere (the drop at $\psi = 0$ is drawn with a small width); the conifold points $\psi^5 = 1$, where the Jacobian ring degenerates, are marked on the unit circle.

VII. THE FERMAT POINT

At $\psi = 0$ the Hodge norms are explicit.

Proposition 9 (Hodge norms at the Fermat point)**.** *Let X_0 be the Fermat quintic and $\omega_{\mathbf{b}}$ the class of $x^{\mathbf{b}}$, $\mathbf{b} \in B$. Write $k_i = b_i + 1$ and $\gamma(t) = \Gamma(t)/\Gamma(1-t)$. There is a constant $C > 0$ independent of $\mathbf{b}$ such that*

$$g_{\mathbf{b}} = \langle \omega_{\mathbf{b}}, \overline{\omega_{\mathbf{b}}} \rangle = C \prod_{i=1}^{5} \gamma(k_i/5). \tag{6}$$

Proof. The full group μ_5^5/μ_5 of diagonal phases acts on X_0, and $\omega_{\mathbf{b}}$ spans the eigenline of the character $\mathbf{k} = (k_i)$, with $1 \le k_i \le 4$ and $\sum k_i = 10$. Eigenlines of distinct characters are orthogonal for the Hodge metric, so $g_{\mathbf{b}}$ is the only entry of the metric on that line, and it is computed from the periods of $\omega_{\mathbf{b}}$ by the second Riemann bilinear relation: if P is an integral cycle and $P_{\mathbf{k}}$ its component in the eigenspace of $\mathbf{k}$ in $H_3(X_0, \mathbb{C})$, then $g_{\mathbf{b}} = |\int_P \omega_{\mathbf{b}}|^2 / |P_{\mathbf{k}} \cdot P_{-\mathbf{k}}|$ up to a factor depending only on the Hodge type. For the cycle of Pham, the join of the five sets of fifth roots of unity, the period is a product of beta integrals, and the intersection number is a product of factors $1 - \zeta^{k_i}$; combining them with $\Gamma(t)\Gamma(1-t) = \pi/\sin \pi t$ gives (6). This is the computation of Candelas [4] on the Fermat quintic and, in the language of the Gepner model, of Distler and Greene [16]; the steps are written out in Appendix C. □

The proposition fixes the ratios of the Hodge norms inside $H^{2,1}$, which is all that the masses of Theorem 3 need. Since $\gamma(1-t) = 1/\gamma(t)$, every norm is a monomial in the two numbers $\gamma(1/5) = 3.94325$ and $\gamma(2/5) = 1.48951$. For the singlet, $\mathbf{k} = (2,2,2,2,2)$ and $g_{\mathbf{1}} = C\gamma(2/5)^5$; for the type $\{2,1,1,1,0\}$, $\mathbf{k}$ is a permutation of $(3,2,2,2,1)$ and $g_{\mathbf{b}} = C\gamma(2/5)^2\gamma(1/5)$; for the type $\{2,2,1,0,0\}$, $\mathbf{k}$ is a permutation of $(3,3,2,1,1)$ and $g_{\mathbf{b}} = C\gamma(1/5)^2/\gamma(2/5)$. The ratio of the second to the first is $\Gamma(3/5)^3\Gamma(1/5)/\Gamma(2/5)^3\Gamma(4/5) = 1.19324$, the square of the constant $\kappa = 1.09236$ in which Distler and Greene [16] expressed the normalised couplings of the Gepner model (see [15, eq. (4.3)]). The norm of Ω, the class of 1 with $\mathbf{k} = (1,1,1,1,1)$, has the same shape with a different constant, and the Weil–Petersson metric of the pencil at the Fermat point, computed from the periods by Candelas, de la Ossa, Green and Parkes [5], is

$$G_{\psi\bar{\psi}}(0) = 25\,\frac{\Gamma(2/5)^5\,\Gamma(4/5)^5}{\Gamma(1/5)^5\,\Gamma(3/5)^5} = 0.19226,$$

the value that Butbaia and collaborators [15] reproduce with numerical Ricci-flat metrics.[5]

Inserting (6) into Theorem 3 gives the fifty-one nonzero masses at the Fermat point. Each label of the types $\{2,1,1,1,0\}$ and $\{2,2,1,0,0\}$ is paired with a label of the same type and the same multiset of k_i, so $g_{\mathbf{b}} = g_{\sigma\mathbf{b}}$ and its mass is $v/g_{\mathbf{b}}$; the singlet has mass $v/g_{\mathbf{1}}$. Table II lists the values. In units of the singlet mass the two doublet masses are 0.83805 (twenty labels, ten doublets) and 0.70234 (thirty labels, fifteen doublets). The fifty generations with an exponent

[5] In [15] the value is given for the mirror quintic, whose complex structure moduli space is the ψ-line of the Dwork pencil divided by the symmetry $\psi \mapsto \zeta\psi$. The Weil–Petersson metric is $-\partial\bar{\partial}\log\int\Omega\wedge\overline{\Omega}$, and passing to the quotient by Γ multiplies $\int\Omega\wedge\overline{\Omega}$ by the constant $1/125$, which does not change the metric; the two numbers are therefore the same. Their machine-learned harmonic representatives give 0.1906 against the exact 0.1923.

TABLE II. Hodge norms and masses at the Fermat point, from (6) and Theorem 3. Norms are in units of C and masses in units of v/C; the types with an exponent three are massless at $\psi = 0$.

type	count	$g_{\mathbf{b}}/C$	value	mass
$\{1,1,1,1,1\}$	1	$\gamma(2/5)^5$	7.33179	0.136392
$\{2,1,1,1,0\}$	20	$\gamma(2/5)^2\gamma(1/5)$	8.74859	0.114304
$\{2,2,1,0,0\}$	30	$\gamma(1/5)^2/\gamma(2/5)$	10.4392	0.095793
$\{3,1,1,0,0\}$	30	$\gamma(2/5)^2\gamma(1/5)$	8.74859	0
$\{3,2,0,0,0\}$	20	$\gamma(1/5)^2/\gamma(2/5)$	10.4392	0

three are massless there, and acquire masses proportional to $|\psi|$ (thirty) and $|\psi|^2$ (twenty) as the modulus moves off the Fermat point.

VIII. THE SELECTION CONJECTURE

A. The landscape and its metric

We use the setting of [24]. A heterotic vacuum is a pair (X,V) of a Calabi–Yau threefold X and a slope-polystable holomorphic bundle V with $c_1(V) = 0$ for which $c_2(T_X) - c_2(V)$ is effective; the standard embedding is the case $V = T_X$. The landscape $\mathscr{L}$ is the set of isomorphism classes of vacua. It is the disjoint union of the components $\mathscr{L}_\tau$ of fixed topological type τ, and each component carries the Riemannian metric $g = G^{\mathrm{WP}} \oplus G^V$ made of the Weil–Petersson metric of the complex structures and the L^2 metric of the bundle moduli.[6] Mirror symmetry enters through an involution μ of a union $\mathscr{L}^\mu$ of components, which sends a vacuum to its mirror and is an isometry of g, as [24] records for the Weil–Petersson metric. A vacuum is *viable* when it satisfies the nine conditions (C1)–(C9) of the phenomenological closure theorem of [24]: anomaly cancellation and stability, three net families, a Wilson line breaking to the Standard Model group, three families without chiral exotics, masses for the vectorlike exotics, doublet–triplet splitting, stabilised moduli, an acceptable rank and hierarchy of the Yukawa masses, and consistent supersymmetry breaking.

Conjecture 10 (Geometric Selection Conjecture [24])**.** There is a functional $\mathscr{S} : \mathscr{L} \to [0,\infty)$, constructed from the metric g, such that

(G1) $\mathscr{S}$ is smooth and proper on each component $\mathscr{L}_\tau$;

(G2) $\mathscr{S}$ has a unique global minimum $\mathscr{V}^\star \in \mathscr{L}$;

(G3) $\mathscr{V}^\star$ is viable;

(G4) $\mathscr{S} \circ \mu = \mathscr{S}$ on $\mathscr{L}^\mu$.

[6] By the theorem of Bogomolov, Tian and Todorov the complex structures of X are unobstructed, and the bundle moduli of a stable bundle are unobstructed when $H^2(X, \mathrm{End}_0 V) = 0$. Where a component is an orbifold or has singular points, everything below is applied on its smooth locus, or on orbifold charts; the only tools used are partitions of unity, geodesic balls and cut-off functions, and all three exist in that generality. As in [24] we take each $\mathscr{L}_\tau$ to be a connected manifold of finite dimension with countably many components in all.

B. The functional adds nothing

Theorem 11. *Conjecture 10 holds if and only if there is a viable vacuum $\mathscr{V}^\star$ with $\mu(\mathscr{V}^\star) = \mathscr{V}^\star$ whenever $\mathscr{V}^\star \in \mathscr{L}^\mu$.*

Proof. Suppose $\mathscr{S}$ exists, and let $\mathscr{V}^\star$ be its minimum. It is viable by (G3). If $\mathscr{V}^\star \in \mathscr{L}^\mu$, then $\mathscr{S}(\mu\mathscr{V}^\star) = \mathscr{S}(\mathscr{V}^\star)$ by (G4), so $\mu\mathscr{V}^\star$ is also a global minimum, and $\mu\mathscr{V}^\star = \mathscr{V}^\star$ by (G2).

Conversely, let $\mathscr{V}^\star$ be viable and fixed by μ when μ is defined there, and let $\mathscr{L}_{\tau^\star}$ be its component. The functional is built in three steps.

A proper function on every component. A second countable manifold M has a smooth proper function $\rho : M \to [0,\infty)$: take compact sets $K_1 \subset K_2 \subset \cdots$ with K_j inside the interior of K_{j+1} and union M, a partition of unity (ϕ_j) with ϕ_j supported in the interior of K_{j+1} minus K_{j-1}, and $\rho = \sum_j j\phi_j$; then $\rho \geq j$ outside K_j. Choose such a ρ_τ on every component. The involution μ permutes the components of $\mathscr{L}^\mu$, so on each orbit of components we may replace ρ by $\rho + \rho\circ\mu$, which is smooth, proper (it is at least ρ) and μ-invariant. Call the result ρ.

A quadratic well at $\mathscr{V}^\star$. Choose $r > 0$ smaller than the injectivity radius of g at $\mathscr{V}^\star$ and such that the closed geodesic ball $\bar{B}$ of radius r about $\mathscr{V}^\star$ is compact. On the open ball B the function $f = d_g(\mathscr{V}^\star,\cdot)^2$ is smooth, since it is the squared norm of the inverse of the exponential map. If $\mathscr{V}^\star \in \mathscr{L}^\mu$, then μ is an isometry fixing $\mathscr{V}^\star$, so it preserves B and $f\circ\mu = f$.

Gluing. Let $\chi : [0,\infty) \to [0,1]$ be smooth, equal to 1 on $[0, r^2/4]$ and to 0 on $[r^2/2,\infty)$, and extend $\chi\circ f$ by zero outside B; the extension is smooth because $\chi\circ f$ vanishes on the part of B at distance more than $r/\sqrt{2}$ from $\mathscr{V}^\star$. Put $c = r^2/4$ and

$$\mathscr{S} = (\chi\circ f)\, f + \big(1 - \chi\circ f\big)\,(c + \rho)$$

on $\mathscr{L}_{\tau^\star}$, and $\mathscr{S} = c + \rho$ on every other component. Then $\mathscr{S}$ is smooth and μ-invariant, because every ingredient is. It vanishes at $\mathscr{V}^\star$ and nowhere else: where $\chi\circ f = 1$ it equals f, which vanishes only at $\mathscr{V}^\star$; where $0 < \chi\circ f < 1$ it is a convex combination of $f \geq r^2/4$ and $c + \rho \geq c$, both positive; and where $\chi\circ f = 0$ it equals $c + \rho \geq c$. So $\mathscr{V}^\star$ is the unique global minimum. Finally a sublevel set $\{\mathscr{S} \leq t\}$ of $\mathscr{L}_{\tau^\star}$ lies in $\bar{B} \cup \{\rho \leq t - c\}$, a union of two compact sets, and it is closed, so it is compact; on the other components the sublevel sets of $c + \rho$ are compact. Thus $\mathscr{S}$ satisfies (G1) to (G4). □

Figure 4 draws the functional of the proof on one component.

Remark 12 (Constructed from the metric). The phrase "constructed from the metric" in Conjecture 10 is not a mathematical condition, and the strongest condition it can be read as is invariance under a group of isometries of $(\mathscr{L}, g)$. The proof gives that reading as well: if G is a compact group of isometries containing μ, averaging ρ over G with its Haar measure gives a G-invariant smooth proper function (if $\int \rho(hx)\, dh \leq t$ then $\rho(hx) \leq t$ for some h, so x lies in the compact set $G\cdot\{\rho \leq t\}$), and the well at a G-fixed vacuum is G-invariant. So the conjecture in that reading is equivalent to the existence of a viable vacuum fixed by G. In every reading the functional is determined by the vacuum it selects and contributes nothing else.

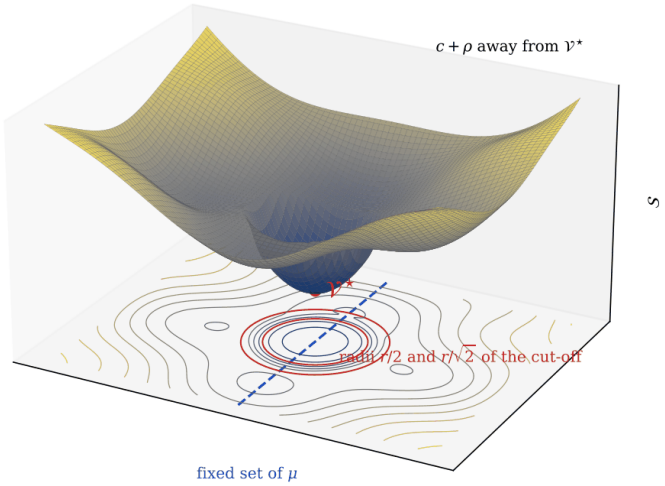


FIG. 4. The functional $\mathscr{S}$ of the proof of Theorem 11 on a two-dimensional component $\mathscr{L}_{\tau^\star}$ on which μ acts by a reflection (dashed line: its fixed set). Inside the radius $r/2$ about $\mathcal{V}^\star$ the functional is the squared distance f; outside the radius $r/\sqrt{2}$ it is $c+\rho$ with ρ proper and μ-invariant; between the two circles it is a convex combination of the two. The contour lines on the floor are the level sets of $\mathscr{S}$; all of them are compact, and the only zero is $\mathcal{V}^\star$.

Remark 13 (Mirror pairs). A viable vacuum outside the standard embedding lies outside $\mathscr{L}^\mu$ when μ is the mirror map of $(2,2)$ vacua, so the condition on μ in Theorem 11 is empty for it. If mirror pairs are identified in $\mathscr{L}$, as one point each, on the ground that they define the same superconformal field theory, the condition is empty for every vacuum and the conjecture is equivalent to the existence of one viable vacuum.

C. The Dwork pencil

Proposition 14. (i) *For $\psi \neq 0$ and $\psi^5 \neq 1$, the mass map of the invariant direction h on X_ψ has rank* 101*, its entries in the basis of labels are* 1*, ψ and ψ^2, on* 51*,* 30 *and* 20 *labels, and the masses of Theorem 3 are $|\psi|^{r(\mathbf{b})}$ times the Hodge factor $(g_{\mathbf{b}} g_{\sigma\mathbf{b}})^{-1/2}$.*

(ii) *Let X be a smooth quintic threefold and H a finite group acting freely on X. The standard embedding on X/H has $100/|H|$ net families. In particular condition* (C2) *fails on X/H for every such H, and no vacuum of the form $(X/H, T_{X/H})$ is viable.*

Proof. Part (i) is Theorem 3. For (ii), a smooth quintic has $h^{1,1} = 1$ and $h^{2,1} = 101$, so $\chi(X) = -200$. A free action multiplies the Euler characteristic of the quotient by $1/|H|$, so $\chi(X/H) = -200/|H|$, and by Theorem 5 the net number of families is $|\chi(X/H)|/2 = 100/|H|$. Wilson lines on X/H break the gauge group but do not change this index. The equation $100/|H| = 3$ has no solution in integers, so (C2) fails. □

The proposition shows where the Dwork pencil stands with respect to the conjecture. The condition on masses is met in the strongest form one can ask of a single modulus: the mass matrix has full rank and three scales 1, $|\psi|$, $|\psi|^2$, which separate as $\psi \to 0$. The pencil fails on the count of families, and it fails there for every free quotient, so no choice of Wilson lines rescues it. Three-family vacua satisfying (C1) to (C4) exist on other threefolds: the heterotic standard models of Braun, He, Ovrut and Pantev [20], of Bouchard and Donagi [21], and the line-bundle models of Anderson, Gray, Lukas and Palti [22, 23], which also address (C5) and (C6). For none of them are (C7), (C8) and (C9) established, and the words "controlled", "acceptable" and "consistent" in those conditions are fixed by comparison with measurement, not by a definition. By Theorem 11 this is the whole of the open part of the Geometric Selection Conjecture.

IX. CONCLUSION

On the Dwork pencil the question of the title has a complete answer. The hundred and one generations are labelled by characters of the symmetry group Γ, each character occurring once; the couplings of three generations are the values of one functional on the socle of the Jacobian ring, equal to a power of the modulus when the symmetry allows them and to zero otherwise; the invariant deformation pairs the generations by an involution with a single fixed point, so its mass matrix is monomial with determinant ψ^{70}; and at the Fermat point the masses are explicit values of the Gamma function. The proofs use two facts only, the reduction (5) and the grading of the Jacobian ring by characters, together with the general dictionary of Section II.

Two questions are not settled by the geometry of the pencil. The breaking of E_6 to the Standard Model needs a free quotient and Wilson lines, and on a quintic the number of fam-

ilies of a free quotient is $100/|H|$, never three (Proposition 14). The selection of the vacuum is reduced by Theorem 11 to the existence of one viable vacuum, and that existence depends on conditions (C7) to (C9), whose thresholds are set by measurement. The same two facts that drive Sections IV to VI are available for the one-parameter families of Fermat type in other weighted projective spaces; whether the characters of their symmetry groups still separate the generations there is a finite computation that this paper does not carry out.

DATA AND CODE AVAILABILITY

The programs that repeat the finite computations of this paper are in its archive [29], which also holds the figures and the record of the engine runs. They are written in C (the label and coupling counts), C++ (the expansion of the Hessian by permutations), Python (all counts, the intersection numbers of Appendix C and the Gamma values), Julia (the Gamma values and the intersection numbers), Macaulay2 (the Jacobian ring at two rational values of ψ, with the socle functional on all 3876 monomials of degree fifteen and the Hessian) and Lean 4 (the counts and the arithmetic of the Hessian, checked by the kernel), with one shell script that runs them all. No proof in the paper depends on them. The constructions were organised on the Anaxia engine, release 3.0.5 [27], described in [28]; its record for the four targets of this paper is part of the archive. The programs and the engine runs were prepared by Deep Bhattacharjee with AI assistance from Claude (Anthropic).

ETHICAL DECLARATIONS

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Appendix A: The Hessian in the socle

The Hessian matrix of F_ψ has entries

$$H_{ii} = 20x_i^3, \qquad H_{ij} = -5\psi \prod_{k\neq i,j} x_k \quad (i \neq j).$$

For a permutation π of $\{1,\dots,5\}$ with m moved points and $f = 5-m$ fixed points, the product $\prod_i H_{i\pi(i)}$ is $20^f(-5\psi)^m$ times a monomial whose exponents are read off as follows. A fixed coordinate a receives 3 from its diagonal entry and 1 from each of the m off-diagonal entries, since an off-diagonal entry $(i,\pi(i))$ omits only the moved coordinates i and $\pi(i)$. A moved coordinate c is omitted by exactly two off-diagonal entries, those with $i = c$ and with $\pi(i) = c$, which are different because $\pi(c) \neq c$; it receives 1 from each of the other $m-2$. Hence the monomial of π is

$$\prod_{a \text{ fixed}} x_a^{3+m} \prod_{c \text{ moved}} x_c^{m-2},$$

of degree $f(3+m)+m(m-2) = 15$, and it depends only on the number of moved points. The five values of m give:

- $m = 0$: the monomial $m_0 = (3,3,3,3,3)$, sector 0;
- $m = 2$: $(5,5,5,0,0)$ up to order, sector 3, so $\lambda = \psi^3$;
- $m = 3$: $(6,6,1,1,1)$ up to order, sector 2, so $\lambda = \psi^2$;
- $m = 4$: $(7,2,2,2,2)$ up to order, sector 1, so $\lambda = \psi$;
- $m = 5$: $(3,3,3,3,3)$, sector 0.

The contribution of a permutation to $H(\psi) = \lambda(\operatorname{Hess} F_\psi)$ is $\operatorname{sgn}(\pi)\,20^f(-5\psi)^m\psi^{r(m)}$ with $r(0) = r(5) = 0$, $r(2) = 3$, $r(3) = 2$, $r(4) = 1$, so $m + r(m) = 5$ for every $m \geq 2$ and every non-identity permutation contributes a multiple of ψ^5. The cycle types are

1^5 : 1 permutation, $m = 0$, $+20^5 = 3\,200\,000$;

$2\,1^3$: 10, odd, $m = 2$: $10\cdot(-1)\cdot 20^3\cdot 25 = -2\,000\,000$;

$3\,1^2$: 20, even, $m = 3$: $20\cdot 20^2\cdot(-125) = -1\,000\,000$;

$4\,1$: 30, odd, $m = 4$: $30\cdot(-1)\cdot 20\cdot 625 = -375\,000$;

$2^2\,1$: 15, even, $m = 4$: $15\cdot 20\cdot 625 = 187\,500$;

5 : 24, even, $m = 5$: $24\cdot(-3125) = -75\,000$;

$3\,2$: 20, odd, $m = 5$: $20\cdot(-1)\cdot(-3125) = 62\,500$,

where the last six numbers are the coefficients of ψ^5. The counts $1+10+20+30+15+24+20$ add up to 120. The coefficients of ψ^5 add up to $-3\,200\,000$, which gives $H(\psi) = 3\,200\,000(1-\psi^5)$.

Appendix B: The coupling count

1. Ordered triples by sector

For $\mathbf{e}$ of degree fifteen, $O(\mathbf{e})$ is the number of ordered triples $(\mathbf{a},\mathbf{b},\mathbf{c}) \in B^3$ with $\mathbf{a}+\mathbf{b}+\mathbf{c} = \mathbf{e}$. We count it by the third label: for each $\mathbf{c} \in B$ with $\mathbf{c} \leq \mathbf{e}$, put $\mathbf{s} = \mathbf{e}-\mathbf{c}$, a vector of degree ten, and count the $\mathbf{a} \in B$ with $\mathbf{s}-\mathbf{a} \in B$, that is, with

$$\max(0, s_i - 3) \leq a_i \leq \min(3, s_i) \quad \text{for all } i, \qquad \sum a_i = 5.$$

This number $N(\mathbf{s})$ is the coefficient of t^5 in $\prod_i \left(t^{\max(0,s_i-3)} + \cdots + t^{\min(3,s_i)}\right)$. Write $[n] = 1 + t + \cdots + t^{n-1}$.

Sector 0, $\mathbf{e} = (3,3,3,3,3)$. Here $\mathbf{s} = 3\mathbf{1} - \mathbf{c}$ has entries in $\{0,\dots,3\}$, so $N(\mathbf{s})$ is the coefficient of t^5 in $\prod_i [s_i+1]$. The multiset of entries of $\mathbf{s}$ is the complement of the type of $\mathbf{c}$:

type of $\mathbf{c}$	entries of $\mathbf{s}$	#$\mathbf{c}$	$N(\mathbf{s})$
$\{3,2,0,0,0\}$	$\{3,3,3,1,0\}$	20	24
$\{3,1,1,0,0\}$	$\{3,3,2,2,0\}$	30	28
$\{2,2,1,0,0\}$	$\{3,3,2,1,1\}$	30	38
$\{2,1,1,1,0\}$	$\{3,2,2,2,1\}$	20	44
$\{1,1,1,1,1\}$	$\{2,2,2,2,2\}$	1	51

The values of N are central coefficients. For $\{3,3,3,1,0\}$, $[4]^3 = 1+3t+6t^2+10t^3+12t^4+12t^5+\cdots$ and the factor $[2]$ gives $12+12 = 24$. For $\{3,3,2,2,0\}$, $[4]^2 = (1,2,3,4,3,2,1)$ and $[3]^2 = (1,2,3,2,1)$, so the coefficient is $2+6+12+$

$6+2=28$. For $\{3,3,2,1,1\}$, $[3][2]^2=(1,3,4,3,1)$ and the coefficient is $2+9+16+9+2=38$. For $\{3,2,2,2,1\}$, $[3]^3=(1,3,6,7,6,3,1)$ and $[4][2]=(1,2,2,2,1)$, so the coefficient is $3+12+14+12+3=44$. For $\{2,2,2,2,2\}$ the coefficient of t^5 in $(1+t+t^2)^5$ is 51. Hence

$$O(3,3,3,3,3)=20\cdot 24+30\cdot 28+30\cdot 38+20\cdot 44+51=3391.$$

(The table for $\mathbf{e}=(3,3,3,3,3)$ also shows that every $\mathbf{c}\in B$ occurs, since $\mathbf{s}=3\mathbf{1}-\mathbf{c}$ always admits some $\mathbf{a}$.)

Sector 1, $\mathbf{e}=(7,2,2,2,2)$. Since $a_1+b_1\le 6$, $c_1\ge 1$, and $c_i\le 2$ for $i\ge 2$. With $\mathbf{s}=\mathbf{e}-\mathbf{c}$, the entry $s_1=7-c_1$ forces $a_1\ge 4-c_1$, and the remaining entries $s_i=2-c_i\le 2$ impose only $a_i\le s_i$. Grouping $\mathbf{c}$ by c_1 and by the multiset of $(c_2,\dots,c_5)$:

c_1	$(c_2,\dots,c_5)$	#**c**	$N(\mathbf{s})$	total
1	$\{1,1,1,1\}$	1	6	6
1	$\{2,1,1,0\}$	12	4	48
1	$\{2,2,0,0\}$	6	3	18
2	$\{1,1,1,0\}$	4	14	56
2	$\{2,1,0,0\}$	12	10	120
3	$\{1,1,0,0\}$	6	26	156
3	$\{2,0,0,0\}$	4	19	76

For instance, when $c_1=3$ and $(c_2,\dots,c_5)=(2,0,0,0)$, a_1 ranges over $\{1,2,3\}$ and the other entries satisfy $a_i\le(0,2,2,2)$ with sum $5-a_1$; the coefficients of t^4, t^3, t^2 in $[3]^3$ are 6, 7, 6, which add to 19. The total is $O(7,2,2,2,2)=480$.

Sector 2, $\mathbf{e}=(6,6,1,1,1)$. Here $c_3,c_4,c_5\le 1$, and the same grouping gives

$\{c_1,c_2\}$	(c_3,c_4,c_5)	#**c**	$N(\mathbf{s})$	total
$\{1,1\}$	$\{1,1,1\}$	1	2	2
$\{2,0\}$	$\{1,1,1\}$	2	1	2
$\{2,1\}$	$\{1,1,0\}$	6	4	24
$\{2,2\}$	$\{1,0,0\}$	3	10	30
$\{3,0\}$	$\{1,1,0\}$	6	2	12
$\{3,1\}$	$\{1,0,0\}$	6	8	48
$\{3,2\}$	$\{0,0,0\}$	2	22	44

and $O(6,6,1,1,1)=162$.

Sector 3, $\mathbf{e}=(5,5,5,0,0)$. Here $c_4=c_5=0$ and $c_1+c_2+c_3=5$. For $\{c_1,c_2,c_3\}=\{2,2,1\}$ the entries of $\mathbf{s}$ are $(3,3,4)$, the bounds are $a_i\le 3$ on the first two and $1\le a_3\le 3$, and the coefficient of t^5 in $[4]^2(t+t^2+t^3)$ is $3+4+3=10$; for $\{3,1,1\}$ the entries are $(2,4,4)$ and the coefficient of t^5 in $[3](t+t^2+t^3)^2$ is $2+3+2=7$; for $\{3,2,0\}$ the entries are $(2,3,5)$ and the coefficient of t^5 in $[3][4](t^2+t^3)$ is $3+3=6$. With three, three and six choices of $\mathbf{c}$, $O(5,5,5,0,0)=30+21+36=87$.

The ordered counts by sector are $O_0=3391$, $O_1=5\cdot 480=2400$, $O_2=10\cdot 162=1620$ and $O_3=10\cdot 87=870$, with total 8281 in agreement with Section V A.

2. Repeated labels

For $\mathbf{a}\in B$ the partner $\mathbf{c}$ with $2\mathbf{a}+\mathbf{c}$ in a sector is found type by type, as in the proof of Theorem 3. For $\mathbf{a}=\mathbf{1}$, $\mathbf{c}=\mathbf{1}$ in sector 0. For $\mathbf{a}$ of type $\{2,1,1,1,0\}$, sector 1 gives $c_i\equiv 2-2a_i$, that is $\mathbf{c}$ with a three where $\mathbf{a}$ has its two, a two where $\mathbf{a}$ has its zero and zeros elsewhere, of sum five. For type $\{3,1,1,0,0\}$, sector 1 gives a one where $\mathbf{a}$ has its three, zeros where it has ones and twos where it has zeros. For type $\{3,2,0,0,0\}$, sector 2 gives zero, two and ones. For type $\{2,2,1,0,0\}$, sector 3 gives ones at the twos, a three at the one and zeros elsewhere. In each case the other sectors produce a residue four or a total different from five. So the hundred triples $\{\mathbf{a},\mathbf{a},\mathbf{c}\}$ with $\mathbf{c}\ne\mathbf{a}$ lie $20+30=50$ in sector 1, 20 in sector 2 and 30 in sector 3, and none in sector 0, which is the list $D_r=0,50,20,30$ used in Section V A.

Appendix C: Hodge norms at the Fermat point

This appendix proves Proposition 9. Write $G=\mu_5^5/\mu_5$ for the group of diagonal phases of $\mathbb{P}^4$, which acts on X_0. For $\mathbf{k}\in\{1,2,3,4\}^5$ with $\sum k_i\equiv 0 \pmod 5$ let $\chi_{\mathbf{k}}(a)=\zeta^{a\cdot\mathbf{k}}$.

1. Eigenlines

The character $\chi_{\mathbf{k}}$ occurs in $H^3(X_0,\mathbb{C})$ exactly once, and these are all the characters that occur: the number of such $\mathbf{k}$ is $\frac{1}{5}\sum_{j=0}^4\big(\sum_{k=1}^4\zeta^{jk}\big)^5=\frac{1}{5}(4^5-4)=204=b_3(X_0)$, and each occurs at least once because the residue $\mathrm{Res}(x^{\mathbf{k}-\mathbf{1}}\Omega_0/F^{q+1})$, $5(q+1)=\sum k_i$, is a nonzero class of character $\chi_{\mathbf{k}}$ (Griffiths' theorem and the fact that $x^{\mathbf{k}-\mathbf{1}}$, with exponents at most three, is nonzero in the Jacobian ring $\mathbb{C}[x]/(x_1^4,\dots,x_5^4)$ of the Fermat quintic). In particular $\omega_{\mathbf{b}}$ spans the line of $\mathbf{k}=\mathbf{b}+\mathbf{1}$.

2. The bilinear relation

Let Q be the intersection form on $H_3(X_0,\mathbb{Z})$, extended to complex coefficients, and identify H^3 with H_3 by Poincaré duality D, so that $\int_c\omega=Q(c,D\omega)$ and $\int_{X_0}\omega\wedge\eta=Q(D\omega,D\eta)$. Since Q is G-invariant, it pairs the eigenline L of χ only with the eigenline L' of $\bar\chi$. Let e be an integral cycle and u its component in L'; its component in L is $\bar u$, because e is real. If $D\omega_{\mathbf{b}}=\alpha\bar u$ with $\bar u\ne 0$, then

$$\int_e\omega_{\mathbf{b}}=Q(e,\alpha\bar u)=\alpha\,Q(u,\bar u),\qquad \int_{X_0}\omega_{\mathbf{b}}\wedge\overline{\omega_{\mathbf{b}}}=|\alpha|^2Q(\bar u,u),$$

and therefore

$$g_{\mathbf{b}}=\Big|\int_{X_0}\omega_{\mathbf{b}}\wedge\overline{\omega_{\mathbf{b}}}\Big|=\frac{|\int_e\omega_{\mathbf{b}}|^2}{|Q(u,\bar u)|}.\tag{C1}$$

It remains to compute both sides for one cycle.

3. The cycle of Pham

On the affine chart $x_5 = 1$ put $y_i = x_i/x_5$; the open part of X_0 is $U = \{f = 0\}$, $f = y_1^5 + y_2^5 + y_3^5 + y_4^5 + 1$, and μ_5^4 acts by $y_i \mapsto \zeta^{a_i} y_i$. A character $\mathbf{k}$ of G restricts to the character (k_1, k_2, k_3, k_4) of μ_5^4, and $k_5 \equiv -(k_1 + k_2 + k_3 + k_4)$. Pham [17] showed that U retracts onto the join of four copies of the set of fifth roots of -1, and that $H_3(U, \mathbb{Z})$ is generated over $\mathbb{Z}[\mu_5^4]$ by the cycle

$$e = (1 - T_1)(1 - T_2)(1 - T_3)(1 - T_4)\,\Delta,$$

where Δ is the real simplex $\{y_i = \varepsilon v_i,\ v_i \geq 0,\ \sum v_i^5 = 1\}$ with $\varepsilon^5 = -1$, and T_i multiplies y_i by ζ.

The period. By Griffiths' residue formula, $\int_e \omega_{\mathbf{b}} = \frac{1}{2\pi i}\int_{\tau(e)} x^{\mathbf{b}}\Omega_0/F^2$ with τ the tube map. On the chart this is the integral of $y^{\mathbf{b}'}dy/f^2$, $\mathbf{b}' = (b_1, \ldots, b_4)$. Put $f_s = \sum y_i^5 + s$; then $y^{\mathbf{b}'}dy/f_s^2 = -\partial_s(y^{\mathbf{b}'}dy/f_s)$, and $\frac{1}{2\pi i}\int_{\tau(e)} y^{\mathbf{b}'}dy/f_s = \int_{e_s} y^{\mathbf{b}'}dy/df_s$ with the Gelfand–Leray form. The substitution $y = s^{1/5}u$ shows that the last integral is $s^A P$ with $A = \frac{1}{5}\sum_{i\leq 4} k_i - 1$ and P its value at $s = 1$. Differentiating at $s = 1$,

$$\int_e \omega_{\mathbf{b}} = -A\,P.$$

On Δ, with $t_i = v_i^5$, the Gelfand–Leray form is $5^{-4}\prod_i t_i^{1/5-1}\,dt$ on the simplex $\sum t_i = 1$, and $y^{\mathbf{b}'} = \varepsilon^{|\mathbf{b}'|}\prod t_i^{b_i/5}$; Dirichlet's integral gives $\int_\Delta = \varepsilon^{|\mathbf{b}'|+4} 5^{-4}\prod_{i\leq 4}\Gamma(k_i/5)/\Gamma(\sum_{i\leq 4} k_i/5)$, and each factor $1 - T_i$ multiplies it by $1 - \zeta^{k_i}$. Since $\sum_{i\leq 4} k_i = 10 - k_5$, $A = 1 - k_5/5$ and $\Gamma(A+1) = A\,\Gamma(A)$, so

$$\Big|\int_e \omega_{\mathbf{b}}\Big| = 5^{-4}\prod_{i=1}^{4}\big|1 - \zeta^{k_i}\big|\,\frac{\prod_{i=1}^4\Gamma(k_i/5)}{\Gamma(1 - k_5/5)}. \tag{C2}$$

The intersection number. For one variable, the reduced homology of the fibre $\{y^5 = -1\}$ is the augmentation ideal I of $\mathbb{Z}[t]/(t^5 - 1)$, the monodromy is multiplication by t, and the variation operator is multiplication by $t - 1$. The Seifert form $L(x, z) = \langle x, \mathrm{Var}^{-1} z\rangle$ is therefore, in terms of the values $\hat{x}(k) = x(\zeta^k)$,

$$L(x, z) = \frac{1}{5}\sum_{k=1}^{4}\frac{\hat{x}(k)\,\hat{z}(-k)}{\zeta^{-k} - 1}.$$

By the theorem of Sebastiani and Thom [18], in the form of Sakamoto [19] for Seifert forms, the Seifert form of $y_1^5 + \cdots + y_4^5$ is, up to sign, the tensor product of four copies, and the intersection form on the three-dimensional fibre is $Q = -L + L^T$. On the eigenline of $(k_1, \ldots, k_4)$ this gives

$$Q(x, z) = \pm\frac{1}{5^4}\,\hat{x}(\mathbf{k})\hat{z}(-\mathbf{k})\Big(\prod_{i\leq 4}\frac{1}{\zeta^{k_i} - 1} - \prod_{i\leq 4}\frac{1}{\zeta^{-k_i} - 1}\Big).$$

With four factors, $\prod(\zeta^{k_i} - 1) = \zeta^{\sum k_i}\prod(\zeta^{-k_i} - 1)$, so the bracket has absolute value $|1 - \zeta^{k_1+k_2+k_3+k_4}|/\prod_{i\leq 4}|1 - \zeta^{k_i}|^2$, and $|1 - \zeta^{k_1+\cdots+k_4}| = |1 - \zeta^{k_5}|$. The cycle e has $\hat{e}(\mathbf{k}) = \prod_{i\leq 4}(1 - \zeta^{k_i})$ up to a unimodular factor, so its component u satisfies

$$|Q(u, \bar{u})| = 5^{-4}\prod_{i=1}^{5}\big|1 - \zeta^{k_i}\big|. \tag{C3}$$

This is nonzero, so $\bar{u} \neq 0$, as needed in (C1).

4. Conclusion of the proof

Inserting (C2) and (C3) in (C1),

$$g_{\mathbf{b}} = 5^{-4}\prod_{i=1}^{4}\big|1 - \zeta^{k_i}\big|\Gamma(k_i/5)^2 \cdot \frac{1}{|1 - \zeta^{k_5}|\,\Gamma(1 - k_5/5)^2}.$$

With $|1 - \zeta^k| = 2\sin(\pi k/5)$ and $\Gamma(t)\Gamma(1-t) = \pi/\sin\pi t$,

$$2\sin(\pi t)\,\Gamma(t)^2 = 2\pi\,\gamma(t), \qquad \frac{1}{2\sin(\pi t)\,\Gamma(1-t)^2} = \frac{\gamma(t)}{2\pi},$$

so $g_{\mathbf{b}} = 5^{-4}(2\pi)^3\prod_{i=1}^5\gamma(k_i/5)$, which is (6) with $C = (2\pi)^3/5^4$. The class of Ω, with $\mathbf{k} = \mathbf{1}$, has no derivative in the period computation, so $|\int_e \Omega| = |P|$ with $\Gamma(\sum_{i\leq 4} k_i/5) = \Gamma(4/5) = \Gamma(1 - k_5/5)$, and the same formula holds: $\|\Omega\|^2 = C\gamma(1/5)^5$. Since $\partial_\psi \mathrm{Res}(\Omega_0/F_\psi) = \mathrm{Res}(5x_1\cdots x_5\Omega_0/F_\psi^2) = 5\omega_{\mathbf{1}}$ and $\omega_{\mathbf{1}}$ is of type $(2,1)$ at $\psi = 0$ (its character differs from that of Ω), the Weil–Petersson metric at the Fermat point is

$$\begin{aligned} G_{\psi\bar{\psi}}(0) &= \frac{25\,g_{\mathbf{1}}}{\|\Omega\|^2} = 25\,\frac{\gamma(2/5)^5}{\gamma(1/5)^5} \\ &= 25\,\frac{\Gamma(2/5)^5\Gamma(4/5)^5}{\Gamma(1/5)^5\Gamma(3/5)^5} = 0.19226. \end{aligned}$$

5. Numerical values

With $\Gamma(1/5) = 4.59084$, $\Gamma(2/5) = 2.21816$, $\Gamma(3/5) = 1.48919$ and $\Gamma(4/5) = 1.16423$, $\gamma(1/5) = 3.94325$ and $\gamma(2/5) = 1.48951$; $\gamma(3/5)$ and $\gamma(4/5)$ are their inverses. The norms of the five types follow by multiplying the five factors, and the masses of Table II are their inverses.

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